

CCA Project

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1 Resultant

For the entire section, let $f, g \in \mathbb{K}[x]$, where $\mathbb{K}$ is a field, with $\deg f = n$ and $\deg g = m$.

First, we define the Sylvester matrix, $S(f, g) \in \mathbb{K}^{(n+m) \times (n+m)}$, as follows [1] :

$$S(f, g) = \begin{pmatrix} f_n & & & g_m & & & \\ f_{n-1} & f_n & & g_{m-1} & g_m & & \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \\ f_0 & f_1 & \cdots & f_n & g_0 & g_1 & \cdots \\ & f_0 & \cdots & \vdots & & g_0 & \cdots \\ & & \ddots & f_1 & & & \ddots \\ & & & f_0 & & & g_0 \end{pmatrix}$$

where all the non-specified coefficients are 0. We can observe that the first m columns of $S(f, g)$ are coefficients of $f(x)$ shifted m times, and the last n columns are coefficients of $g(x)$ shifted n times. We can also define the matrix with reverse order of f and g , or by shifting in rows instead of columns.

Then, the resultant of f and g is defined as [1]:

$$\text{res}(f, g) = \det S(f, g)$$

Note that this definition of the resultant does not change if we define the Sylvester matrix over rows instead of columns, as this is just the transpose of the Sylvester matrix and determinant stays the same on the matrix transpose. If coefficients are in the reverse order, then it means that there are $n + m$ row permutations (or column permutations in the row-oriented definition), thus the determinant changes sign with $(-1)^{n+m}$.

The resultant is important, as we have the following property:

$$\text{res}(f, g) \neq 0 \iff \gcd(f, g) = 1$$

which implies that the resultant of two polynomials is 0 if and only if there is a non-constant polynomial in $\mathbb{K}$ that divides both f and g . This can be interpreted as that f and g have a common root in $\overline{\mathbb{K}}$, the algebraic closure of $\mathbb{K}$: This final point is important since if $\mathbb{K}$ is not algebraically closed, then the common divisor of f and g may have all of its roots in the algebraic closure of $\mathbb{K}$, so that f and g do not share any common roots in $\mathbb{K}$.

Proof. Let $\mathbb{K}[x]_{<d}$ be the $\mathbb{K}$ -vector space with polynomials of degree $< d$, and let its standard bases $(1, x, \dots, x^{d-1})$.

First, let us reexamine the definition of the Sylvester matrix: We define the $\mathbb{K}$ linear map:

$$\phi : \mathbb{K}[x]_{<m} \oplus \mathbb{K}[x]_{<n} \rightarrow \mathbb{K}[x]_{<n+m}$$

such that:

$$\phi(u, v) = uf + vg$$

This definition is a linear map since the addition of polynomials and multiplication of polynomials with a constant polynomial is linear.

Then, we can observe that the matrix of ϕ in the standard bases, $\text{Mat}(\phi)$, is exactly the Sylvester matrix, $S(f, g)$, for a row-oriented definition.¹ Hence,

$$\text{res}(f, g) = 0 \iff \det(S(f, g)) = 0 \iff \det(\text{Mat}(\phi)) = 0 \iff \ker(\text{Mat}(\phi)) \neq \emptyset$$

¹Again, the definition of the Sylvester matrix does not change if resultant is different than 0, thus without loss of generality we can prove the above equivalency for any definition of the Sylvester matrix.

As a result, we can say that $\text{res}(f, g)$ equals 0 if and only if $\ker(\text{Mat}(\phi))$ is not empty, meaning that there exists two vectors not both zero such that their value in ϕ equals 0. Formally:

$$\text{res}(f, g) = 0 \iff \exists(u, v) \neq (0, 0), \deg(u) < m, \deg(v) < n : uf + vg = 0$$

We show the contraposition, $\gcd(f, g) \neq 1 \iff \text{res}(f, g) = 0$.

($\implies$): Assume $\gcd(f, g) \neq 1$. Let $h = \gcd(f, g)$, that is h is non-constant and divides f and g . Let $f_1, g_1 \in \mathbb{K}[x]$ such that:

$$f = hf_1, g = hg_1$$

Then, we set $u = g_1$ and $v = -f_1$, so that:

$$uf + vg = g_1hf_1 - f_1hg_1 = h \times (g_1f_1 - f_1g_1) = 0$$

As $\deg(h) > 0$, we have $\deg(u) < m, \deg(v) < n$. Thus, by the above equality,

$$\gcd(f, g) \neq 1 \implies \exists(u, v) \neq (0, 0), \deg(u) < m, \deg(v) < n : uf + vg = 0 \implies \text{res}(f, g) = 0$$

Simply, polynomials g_1 and $-f_1$ are polynomials that satisfy the above definition so that $\text{res}(f, g) = 0$.

($\impliedby$): Assume $\text{res}(f, g) = 0$. Then, by the above equality we have:

$$(\exists(u, v) \neq (0, 0), \deg(u) < m, \deg(v) < n : uf + vg = 0) \implies uf = -vg$$

We know $u \neq 0$, as g is not 0 and u and v cannot be both 0. Let $d = \gcd(u, g)$ such that:

$$u = du_1, g = dg_1 \text{ where } \gcd(u_1, g_1) = 1$$

Putting this on the above equation:

$$\begin{aligned} (du_1) \times f &= -v \times (dg_1) \implies u_1f = -vg_1 \\ \implies g_1 \mid u_1f &\implies g_1 \mid f \text{ since } \gcd(u_1, g_1) = 1 \end{aligned}$$

Thus, we now have g_1 as the common factor of g and f . Now showing that g_1 is not constant. We know $\deg(u) < m = \deg(g)$:

$$\deg(u) < \deg(g) \implies \deg(du_1) < \deg(dg_1)$$

$$\implies \deg(d) + \deg(u_1) < \deg(d) + \deg(g_1) \implies \deg(u_1) < \deg(g_1)$$

Since we have $\deg(u_1) \geq 0, \deg(g_1) > 0$. This shows that g_1 is a non-constant common divisor of f and g , so $\gcd(f, g) \neq 1$.

We conclude that

$$\text{res}(f, g) \neq 0 \iff \gcd(f, g) = 1$$

□

2 Subresultants

As before, for the entire section, let $f, g \in \mathbb{Z}[x]$ with $\deg f = n$ and $\deg g = m$. Moreover, we assume $m \leq n$ without loss of generality.

Resultant of polynomials is important, as it gives us an efficient manner to understand if two polynomials have a common root or not. However, it does not give any other information but about gcd. Thus, we can extend the resultant definition into subresultants, which include all results of the Extended Euclidean Algorithm [2].

Specifically, the resultant is directly related to the GCD, as shown in the previous section, that it disappears whenever the GCD is constant. In the case of subresultant, the subresultant polynomial (i.e. the polynomial that is constructed through subresultant determinants, explained below) expresses the remainders (with a scaling factor) in the sequence of the euclidean GCD algorithm. The advantage over the regular algorithm is that in the case of subresultant polynomials, as they are defined through the determinant of a matrix in $\mathbb{Z}$, they are in $\mathbb{Z}[x]$, which prevents coefficients from growing too big. This is why it is important to be able to compute subresultant polynomials efficiently.

First, we define Sylvester matrices for subresultants, similar to the resultant's matrix. Given $0 \leq k \leq m$,

$$S_k(f, g) = \begin{pmatrix} f_n & & & g_m & & & \\ f_{n-1} & f_n & & g_{m-1} & g_m & & \\ \vdots & & \ddots & \vdots & & \ddots & \\ f_{n-m+k+1} & \cdots & \cdots & f_n & g_{k+1} & \cdots & \cdots & g_m \\ \vdots & & & \vdots & \vdots & & & \ddots \\ f_{k+1} & \cdots & \cdots & f_m & g_{m-n+k+1} & \cdots & \cdots & \cdots & g_m \\ \vdots & & & \vdots & \vdots & & & & \vdots \\ f_{2k-m+1} & \cdots & \cdots & f_k & g_{2k-n+1} & \cdots & \cdots & \cdots & g_k \end{pmatrix}$$

, which can be understood as removing the last k -columns of each submatrix of f and g , and then removing the last $2k$ rows of the matrix.

Then, the k -th subresultant of f and g is defined as [2]:

$$\alpha_k = \det S_k(f, g)$$

Finally, the subresultant matrix is the determinant of a matrix that equals the multiplication of the subresultant Sylvester matrix with the matrix encoding the basis $(1, x, \dots, x^k)$.

3 Multivariate Polynomials

Let $k > 1$ and $f, g \in \mathbb{Z}[x_1, \dots, x_k] = \mathbb{Z}[x_2, \dots, x_k][x_1]$, with $\deg f = n$ and $\deg g = m$. Then, by the definition of the resultant as the determinant of the Sylvester matrix we know that $\text{Res}_{x_1}(f, g) \in \mathbb{Z}[x_2, \dots, x_k]$. Moreover, we can write that:

$$\deg(\text{Res}_{x_1}(f, g)) \leq \deg_{x_1}(g) \times n + \deg_{x_1}(f) \times m$$

Informally, this results can be deduced from the fact that determinant is defined as the sum over all permutations of n of multiplication of coefficients in the Sylvester matrix. The coefficients of Sylvester matrix are either 0, a coefficient of f , or g . Thus, we can upper bound the degree of each term with the degree of the multiplication on each term. Here, an important point is that we shift coefficients of f exactly m times while coefficients of g n times. Thus, terms belonging to f (this is $\deg(f) = n$) are multiplied with the degree of g over x_1 , which is

$\deg_{x_1}(g)$, and vice versa for g . Finally, we sum both degrees to express the upper bound of the multiplication of all terms, which is the degree of the biggest term on the left (depending on f) times the degree of the biggest term on the right (depending on g).

Then, let $\eta \in \mathbb{Z}^{n-1}$, $f(\eta), g(\eta) \in \mathbb{Z}[x_1]$. Then, we cannot say that the below holds for any chosen η :

$$\text{Res}_{x_1}(f(\eta), g(\eta)) = \text{Res}_{x_1}(f, g)(\eta)$$

In other terms, if $O \subseteq \mathbb{Z}^{n-1}$ such that $\forall \eta \in O : \text{Res}_{x_1}(f(\eta), g(\eta)) = \text{Res}_{x_1}(f, g)(\eta)$, then $O \neq \mathbb{Z}^{n-1}$.

Normally, this equation seems correct, as the determinant of the Sylvester matrix is a polynomial expression in coefficients, so the evaluation commutes. However, resultant is defined as the determinant of the Sylvester matrix. Thus, in order to understand for what values of η the above equality does not hold, we need to inspect the construction of the Sylvester matrix.

As said before, each term of the Sylvester matrix is a polynomial expression, thus the order of evaluation does not change the value of a specific term. However, the determinant actually changes if the matrix inside has a different size. This is the most important observation on this phenomenon: It is not possible for an individual term of the matrix to change based on the order of the evaluation, but the actual size of the matrix may change, which happens if and only if the degree of f or g changes (as the size of the Sylvester matrix depends on the sizes of f and g only by construction).

In short, we can say that the definition of the Sylvester matrix is not the same for the 2 sides of the equation: If the resultant is computed over f, g , then the matrix has a generic size of that depends on $\deg(f)$ and $\deg(g)$. However, if we first evaluate the matrix, then the size of the matrix depends on $\deg(f(\eta))$ and $\deg(g(\eta))$. Formally:

$$\text{Res}_{x_1}(f(\eta), g(\eta)) \neq \text{Res}_{x_1}(f, g)(\eta) \iff \deg(f)! = \deg(f(\eta)) \text{ or } \deg(g)! = \deg(g(\eta))$$

As a result, we need to find values of η that changes the degree of f or g . Trivially, these are the values of η that evaluate the leading coefficient of f or g 0. Formally, we can express the values of η that does not satisfy this equality as:

$$\mathbb{Z}^{n-1} \setminus O = \{\eta \in \mathbb{Z}^{n-1} : lc(f(\eta)) = 0 \text{ or } lc(g(\eta)) = 0\}$$

Now that we express the difference in this form, we can observe that this expression is actually inside the union of two Zariski-closed sets in a field $\mathbb{K}^{n-1}$ (that are expressed with V)².

$$\{\eta \in \mathbb{Z}^{n-1} : lc(f(\eta)) = 0 \text{ or } lc(g(\eta)) = 0\} \subseteq \{\eta \in \mathbb{K}^{n-1} : lc(f(\eta)) = 0 \text{ or } lc(g(\eta)) = 0\}$$

where

$$\{\eta \in \mathbb{K}^{n-1} : lc(f(\eta)) = 0 \text{ or } lc(g(\eta)) = 0\} = V(lc(f)) \cup V(lc(g)) = V(lc(f)lc(g))$$

In short, we can say that

$$\mathbb{Z}^{n-1} \setminus O \subseteq V(lc(f)lc(g))$$

Finally, as $\mathbb{Z}^{n-1} \setminus O$ is included in a closed set, the equation $\text{Res}_{x_1}(f(\eta), g(\eta)) = \text{Res}_{x_1}(f, g)(\eta)$ is generically true, i.e. it holds for all values of η except on a Zariski-closed set.

²Note that since we define Zariski-closed sets on fields, we express this equality as an inclusion: There are values of η that evaluate f or g 0 but does not live in $\mathbb{Z}$

References

- [1] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra*. Cambridge University Press, Cambridge, 1999. Section 6.3: The resultant.
- [2] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra*. Cambridge University Press, Cambridge, 1999. Section 6.10: Subresultants.